MAPPING THE VIOLENT SKY

The scientists who discovered the solution to the cosmic age crisis in the late 1990s weren't trying to revolutionize physics. They were trying to answer a seemingly straightforward question: how quickly is the expansion of the universe *slowing down*? It was common knowledge at the time that the expansion of the cosmos was set off by the Big Bang, and that the gravity of everything inside it has been slowing it down ever since. Measuring one number—the so-called deceleration parameter—would tell us the balance between the outward momentum from the Big Bang and the inward pull of the gravity of everything the universe is made of. The higher the deceleration parameter, the harder gravity is pushing the brakes on cosmic expansion. A high number would indicate the universe is fated for a Big Crunch; a low one would suggest that even though the expansion is slowing, it will never completely stop.

Of course, to measure deceleration, you have to find a way to measure how quickly the universe was expanding in the past, and compare that to how quickly it's expanding now. Fortunately, that whole thing where we can see the past directly by looking at distant things, coupled with the bit where the expansion of the universe makes everything look like it's moving away from us, means that this is totally possible. All we have to do is look at something nearby, and something really far away, see how quickly they're each moving away from us, and apply a little math. Simple!

Okay, in practice it's not simple at all, because you have to know the distances as well as the redshifts, and distances are hard to measure across deep space. But suffice to say, the measurement is *possible*, if very, very difficult. Fortunately, astronomers have a vast and varied toolkit for measuring things in the cosmos, and in this case it turns out that cataclysmic thermonuclear explosions of distant stars do just the trick!

The short explanation is that certain types of supernovae make explosions whose properties are so predictable we can use them as mile-markers for the universe. They involve the violent deaths of white dwarf stars, which are, when they're not busy exploding, the kind of slowly cooling stellar remnant that our Sun will eventually become after it gets through its planet-murdering red giant phase. When a white dwarf grows to a certain critical mass (either by pulling matter off a companion star or by colliding with another white dwarf),[fn1] it detonates. This is called a Type Ia